

Ecal-P Test Beam 2025 Analysis

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1 Introduction

The following paper presents the results and conclusions from my work with the Ecal-p group as part of the Tel Aviv University analysis team, under the LUXE experiment in DESY.

1.1 LUXE Mission and Experimental Methodology

The LUXE experiment is proposed to probe quantum electrodynamics in the non perturbative regime (NP-QED). The heart of the experiment is collision of high energy electrons and photons, where the photons are created by a strong laser. The interaction with a strong laser can also be described as interaction with a very strong electromagnetic (EM) field, thus reaching the non perturbative regime. Further information about the experiment goals and the theory behind it can be found in the LUXE TDR [1].

1.2 Ecal-P Detectors

When performing the LUXE experiment, it is expected to yield a very large flux of particles, with a large range of energies. this creates a difficulty in detecting the outgoing particles (mostly electrons and positrons) since their overlap makes it complicated to distinguish between them by the signals given from the detector.

The Ecal detectors we build in our group are designed to overcome this challenge as much as possible. In a calorimeter particles undergo an EM shower and deposit their energy during the process. The size of the EM shower in a medium is described by its Moliere Radius - the radius of the cylinder in which 90% of the EM shower is contained [2].

Where large amount of particles enter the detector simultaneously, the ability to distinguish them will be largely affected by the Moliere radius - the smaller it will be, the smaller the particles' showers overlap. The Ecal detectors are made of layers of silicon and tungsten, where the tungsten acts as the absorber and the silicon is the signal collector. The Moliere radius in tungsten is considerably small due to its high atomic number ($Z=74$) [3]. The limit for the Moliere radius is determined by the gaps between the tungsten plates where we insert the silicon - the closer we could stack the layers together, the smaller the radius. At the time of writing this paper, the Moliere radius in our calorimeter is 8 [mm].

1.3 Small Gaps Within the Detector

The Ecal detector is composed of: 20 layers of silicon containing 256 pixels are stacked parallel to each other between tungsten plates of 1 radiation length (XO) width, this entire structure is called a **Tower**. Then towers can be placed one next to each other to increase the spatial range of the detector.

However, placing two silicon plates one next to each other results in having a gap between them, where the signal from the particles cannot be collected, this means that between neighboring towers we have an inevitable blind zone.

Further in this paper I will present the impact of this gaps and their properties.

1.4 Pad Reconstructions

As the Ecal detector contains 256 pads (or pixels) and each one have its own electronics, we result with some of the not working, meaning they can not register any signal. the dead pads can be seen in figure (1)

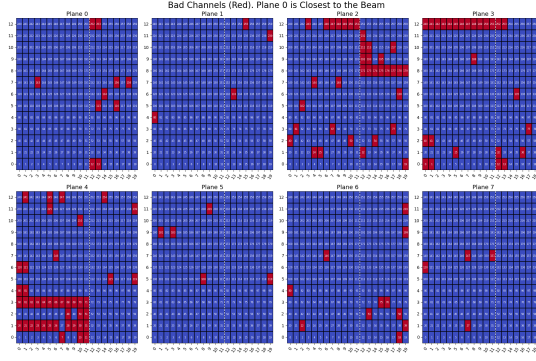


Figure 1: A map of the not working pads in the silicon sensors. Blue are the working pads and Red are the dead ones.

There are a many approaches to reconstruct the signal in a dead pad, and in all work in this paper, I used the Z-average method. This means that to reconstruct the signal in a dead pad in a certain plane, I took the average of the signals in the same pad but in the previous and next planes

$$E_{reco} = \frac{E_+ + E_-}{2} \quad (1)$$

1.5 Telescope Uses and Possible Problems

Before the sensor on the z axis, we have placed a telescope device. This is 6 plates of high granulated silicon sensors. As the particle goes through it, it leaves a signal in the telescope that allows us to determine its trajectory. From the trajectory we have some very useful information. We can determine the coordinate of the hit of a particle on the first plane $\vec{R}_0 = (x_0, y_0, z_0)$ where z_0 is the position on the z axis. We also get the momentum of the particle $\vec{P} = (p_x, p_y, p_z)$.

If we then want the position of the particle at any further plane where $z > z_0$, we simply use the equation:

$$i = \frac{p_i}{p_z}(z - z_0) + i_0, \quad i = x, y \quad (2)$$

In figure(2) each dot is a particle hit on the first plane (z_0) of the sensor, by its given position from the telescope. The telescope enables us to see not only the pad in the sensor that was hit, but also the position inside the pad.

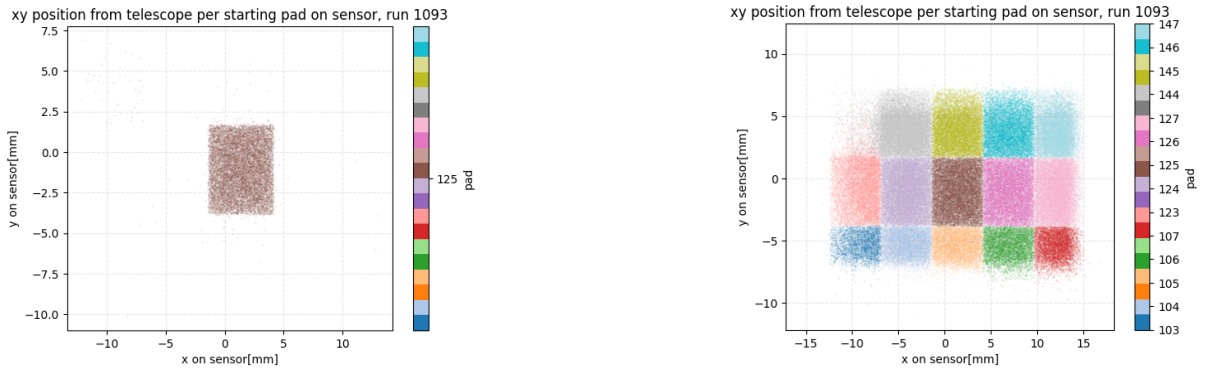


Figure 2: Run 1093: (x, y) distribution for events with a single telescope track and a single hit in the first silicon plane.

However! in some cases the telescope data was distorted. We found that in run 1081, the telescope data becomes less accurate as the time of the run proceeds. While it works fine on the first events, it is not reliable on the last events. See figure (3). The issue in this run has already been addressed and fixed by our data acquisition team.

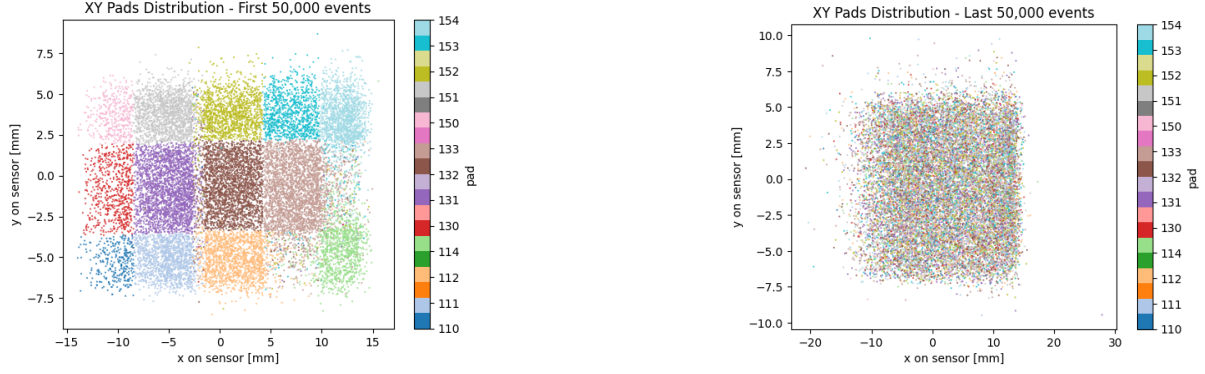


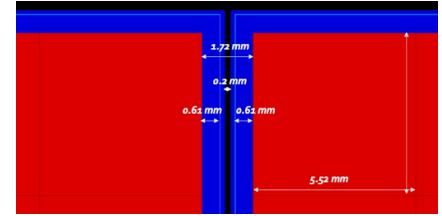
Figure 3: Run 1081: (x, y) position distribution of hits by the telescope. Works well for the first 50k events but gets distorted on the last 50k events.

2 Effective Sensor Geometry Using Tracker Mode

The properties of the sensors are given by the manufacturer - Hamamatsu, as seen in figure(4)

Parameter	Rating	Unit
Device type	P+ PIXEL on N substrate	
Chip size	$89700 \pm 40 \times 89700 \pm 40$	μm
Active area	88480×88480	μm
Chip thickness	320 ± 15	μm
Number of PIXELs	$256(16 \times 16)$	ch
PIXEL pitch	5530×5530	μm
PIXEL GAP	10	μm

(a) Properties of the sensor



(b) Description of the gap between the towers

Figure 4

for this analysis we used a run made in tracker mode, where there are only silicon plates and no tungsten, this means that no EM shower occur when the particle travels through the detector. For all analysis in the section, I used Run 1491 as its located on the te gap between the towers.

My motivation for this was to study the efficiency of the pads at their edge, both between towers (edge of the tower) and between pads (within the tower).

Looking at the position distribution of signal deposition of the events(according to the telescope) we can see in figure (5) that some energy is deposited in a pad even if the particle itself passed outside of the pad's boundaries(the black grid). This means that the pad can collect some energy from particles that are close to him, due to the electric field used in the silicon semiconductor to detect the Ionization. Hence comes the question of what is the effective range of the pad.

2.1 Length of the Pad Edges

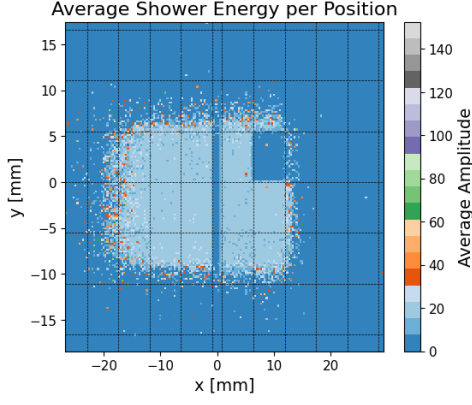
To describe the boundaries of the pad, we take the average signal from all events with the same x value. We assume that on the effective active area of the pad, the signal is homogeneous, thus reading the same signal for equal energy particles, and outside this area we will get a readout of 0. to describe a behaviour from zero to a constant within some range, I find it best to use a sigmoid function.

The sigmoid we used is the logistic function:

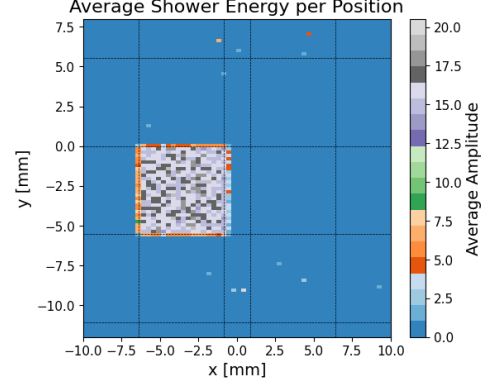
$$f(x) = A \cdot \frac{1}{1 + e^{-r(x-t)}} \quad (3)$$

Where A is the amplitude of the sigmoid and represents the average signal in the active area of the pad, t is the center of the sigmoid where we have 50% of the signal, and r is the rate of the transition from 0 to A .

To determine the length of the pad's edge, we need to determine the sigmoid's length, that depends on how we choose our sigmoid's boundaries. In our case, unless stated otherwise, we will determine this



(a) signal distribution (in all pads)



(b) Signal distribution in pad 91 only. We see the excess signal at the edges of the pad.

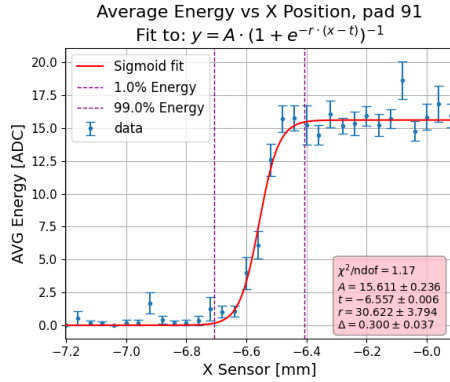
Figure 5: Run 1491: signal distribution in the third plane

boundaries as where we have $E_- = 1\%$ and $E_+ = 99\%$ of the energy. The sigmoid's length will be given by the following equation:

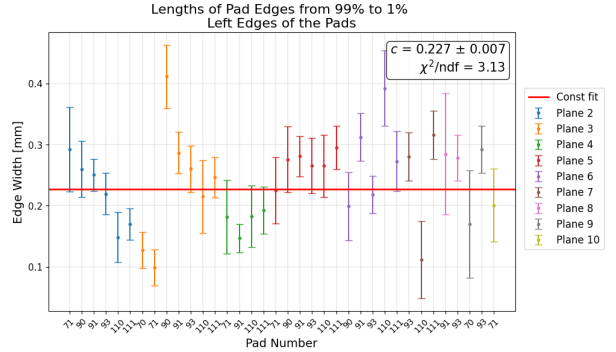
$$\Delta = \frac{2}{r} \cdot \ln\left(\frac{E_+}{E_-}\right) = \frac{2}{r} \ln(99) \quad (4)$$

Hence, the sigmoids length will be the distance in X axis from 1 to 99 percent of the energy. As can be seen as the distance between the purple vertical lines in plot (6a)

In plot (6b) is the length of the left edge for many pads, fitted to a constant value to determine the weighted average of the left edge.



(a) Signal vs X at the left edge of a single pad.



(b) Average effective length of the the left edge of a pad, within the tower.

Figure 6: Run 1491 left-edge signal, fit to a sigmoid. For a single pad and for multiple pads.

As we assessed the average length of the left edge of the pad within the tower, we can repeat this for the right edges within the towers. And also quantify the length of the left and right edges between the towers. As described in figure(7), where blue are edges within towers and red are between towers. The calculated values are presented in table(1). One can see the that the edges of the pads within the tower are smaller than those at the edge of the tower. This is likely due to the electric field in the silicon being distorted at the edge of the sensor where the electronics experience some boundary effect.

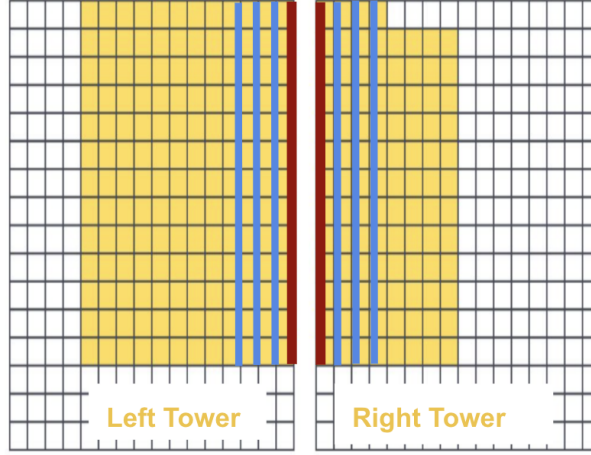


Figure 7: The X edges of the pads in the sensor, blue are edges between pads and red are between towers

	Between Pads (Blue)	Between Towers (Red)
Left Edge [mm]	0.223 ± 0.007	0.34 ± 0.01
Right Edge [mm]	0.23 ± 0.01	0.37 ± 0.01

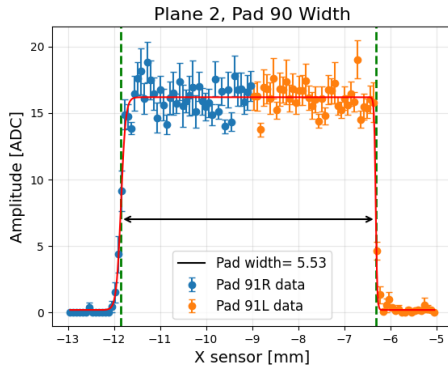
Table 1: Average length [mm] of the left and right edges, both between pads(smaller) and between towers(larger).

2.2 Width of the Pad

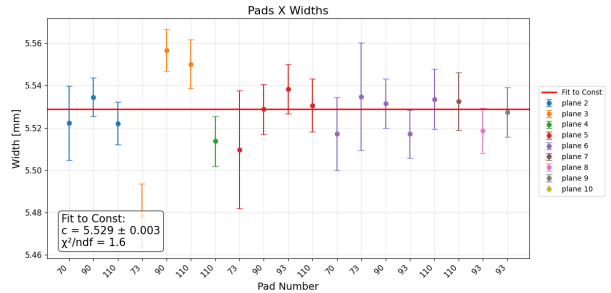
We can also asses the effective width of the pad using the sigmoid fitting. We plot the signal measured in the pad in a range of X covering both of his edges, and we get the signal within the pad and 0 outside of it, then we fit both edges to a sigmoid and find their centers, where the energy is half of what we get inside the pad. Then the distance between the centers of the sigmoids gives us the width of the pad, as shown in plot(8a).

We repeat the calculation of the pad edge for many pads and fit it to a constant to get the wheighed average pad width, seen in plot(8b).

the results in plot(8) gives the pad width **off** the gap, meaning that its calculated for pads that are between other pads. This process was repeated for pads **on** to gap which are at the edge of the tower. The results for the pads widths are shown in table(2)



(a) Assessing the width of a pad as the distance between the centers of his two edges, using a fit to sigmoid.



(b) Taking the width of many pads and fitting to a constant to get the average pad width.

Figure 8: pad width within the the sensor(between pads)

In table(2) one can see that the pad is effectively wider when between towers (at the edge of the tower) than it is when between pads. This makes sense as the edge of the pad is longer when between towers,

	Between Pads (Blue)	Between Towers (Red)	Manufacturer
Effective Pad Size [mm]	5.529 ± 0.003	5.728 ± 0.002	5.53

Table 2: Average pad X width both between pads(smaller) and within towers(larger)

as discussed in section(2.1).

We also see that when between pads, the width aligns with the properties from the manufacturer. The pitch of the sensor is defined as. $5.53[mm]$ meaning that the sensor is symmetric to translations of this distance on the x or y axis. since we calculated the pad width from 0.5 signal on the left to 0.5 on the right, we expect to get exactly the pitch, since the 0.5 signal position on the right of one pad is also the 0.5 signal position on the left of the next pad.

2.3 Gap Size Between Towers

By looking at the edges of pads across different sides of the gap, we can determine where the towers effectively ends, and deduct the Gap size.

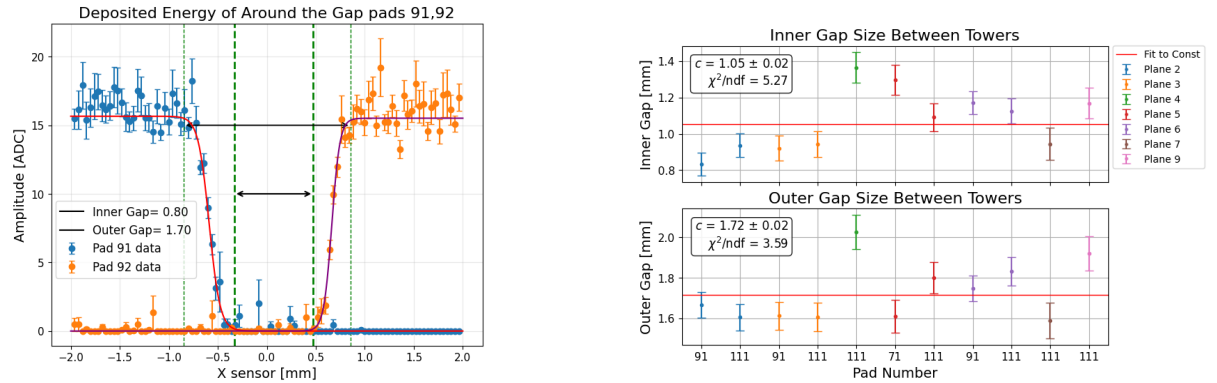
Here we have two measures for the gap:

- Outer Gap - distance between where the pads read 99% of the signal
- Inner Gap - distance between where the pads read 1% of the signal

In plot(9a) we see the effective gap between pads 91 and 92 on the third silicon plate of the detector.

In plot(9b) we have the average inner and outer gap sizes, using a weighted fit to a constant.

The results of this fits are shown in Table(3)



(a) Inner and outer gap sizes between neighboring pads at different sides of the gap

(b) gap sizes for many silicon plates across the detector, fitted to a constant to get the average.

Figure 9

	Outer Gap	Inner Gap	Manufacturer
Effective Gap Size [mm]	1.72 ± 0.02	1.05 ± 0.02	1.72

Table 3: Effective gap sizes, showing the entire gap and the fully blind zone regime.

In the results shown in table(3) we see that the gap size is indeed as given by the manufacturer (4b). BUT, not all the gap is a blind zone, and while we lose the efficiency within it, we get a fully blind zone only at $1.05[mm]$ within the gap, where no signal can be detected.

3 Energy Distribution Around the Gap

In following section presented are the methods and results for quantifying the loss of signal in an event due to the gap.

For this analysis we used runs in configuration of 8 layers of sandwiched tungsten and silicon, where the first layer is tungsten, the electron beam is perpendicular to the detector, and the gap is between columns 11 and 12, marked as the vertical red line seen in plot(10)

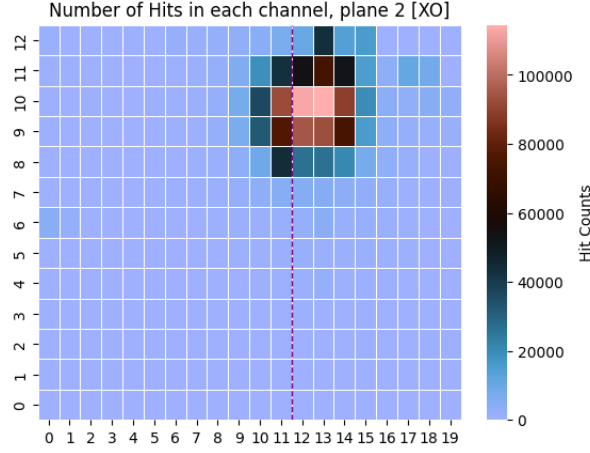


Figure 10: Run 1079: amount of hits per pad in the first plane (after 1 layer of tungsten). Beam located on the upper part of the detector and on the gap

3.1 Evaluating the Signal Loss of a Shower Around the Gap

Lookin at the average signal distributio in the detector, we get that there is a major drop on the vertical line where the gap lies. As seen in plot(11) where we used the telescope to determine where an event started, and calculated the average shower energy for each starting point.

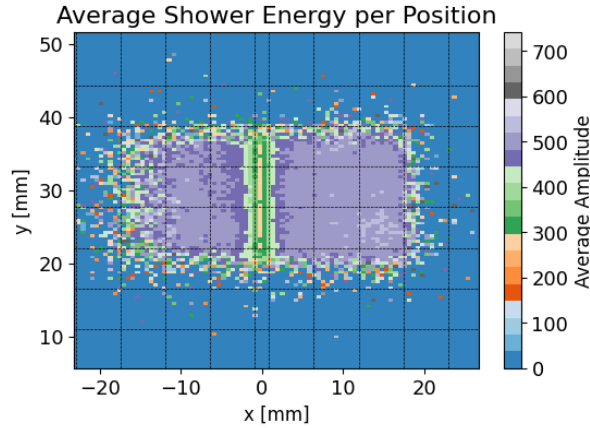


Figure 11: Run 1079: average shower signal distribution in MIP. The green area in the middle shows the signal drop at the gap, clearly dividing the two towers.

Since the gap is on the y axis, we assume that the energy drop depends on the initial x position of the event (but only when the beam and detector are perpendicular!). Thus, we project on the x axis the result from plot(11), and get the average signal loss of a shower vs the initial x of the event, seen in figure(12). We project only data from a slice of y in size 10[mm] from the center to avoid fluctuations due to areas of y with little data, more details are in appendix(??)

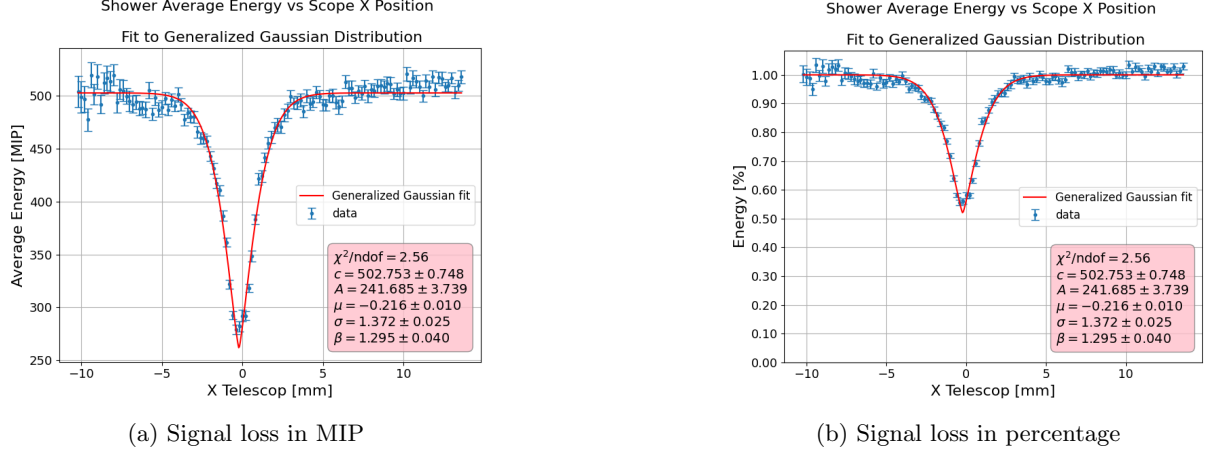


Figure 12: Run 1079: correlation between average shower signal loss and initial x from the telescope. fit to generalized Gaussian function

To quantify the signal loss in plot(12), the data is fitted to a Generalized Gaussian function:

$$f(x) = c - A \cdot e^{-|\frac{x-\mu}{\sigma}|^\beta} \quad (5)$$

Where:

- c - average energy of events not in the gap
- A - the depth of the gaussian, maximal signal loss
- μ - the x position of the gap, and the maximum signal loss
- σ - the spread of the gap
- β - the rate of the Gaussian dip growth

The Generalized Gaussian works great to describe the signal loss because it describes a dip like in a regular gaussian, but allows the modify the width of the dip in different regions, in plot(12) we see that the dip is wider and the start, but more spikey at the maximum signal loss. This make sense when considering the results from section(2.3)

3.2 Gap Profile of Various Runs

We will calculate the gap charactharistics as shown in section(3.1) for multiple runs positioned around the gap with 8 layers of tungsten to study the signal loss.

In table

Table 4: Summary of the gap signal-loss measurements for runs with 8 tungsten layers.

Run Number	Max Signal Loss [MIP]	Avg Energy [MIP]	Signal Loss [%]
1079	242 ± 4	502 ± 1	48 ± 4
1080	238 ± 4	511 ± 1	47 ± 4
1088	226 ± 4	495 ± 1	46 ± 4
1089	246 ± 4	493 ± 1	50 ± 4

Table 5: Maximal signal loss in MIP and in percentage compared to the average signal readout from a shower outside the gap. The results are shown in appendix(5)

We see in the results from table(4) that the signal loss can reach up to 50% of the signal given from a shower. As discussed in section(1.2) the Moliere radius of a shower in our detector is around $8[mm]$ meaning that a gap of $1.72[mm]$ is relatively big! Our compact calorimeter has made the affect of the

gaps more significant. However, having compact showers means that they need to be very close to the gap to lose signal, as seen in figure(12) that the loss is 0 for events starting more than 8[mm] away from the gap, which is less than 2 pads away!

4 Appendix: Gap Profile Fluctuations Over the y Axis

In this appendix we will see how the gap values depends on the initial y value of the event. To do so, we will generate the energy vs x plot for different segments of y, and extract the parameters c, A, μ and σ and understand how they vary along the y axis.

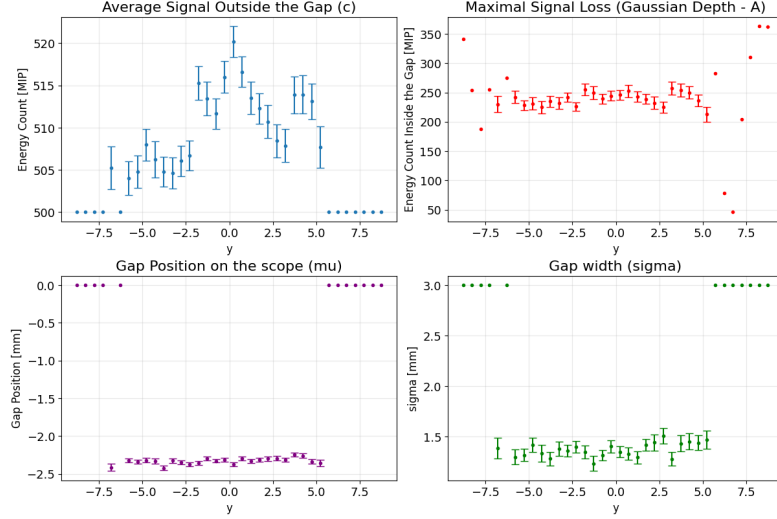


Figure 13: Run 1088: fitting parameters of the gap for varying range of y. $-9 < y < 9$, with segments of 0.5[mm]. We see here the uncertainty rise due to poor statistics as we get closer to the boundaries of the shower where $|y| > 5$.

We will calculate the parameters of the gap for range of $-5 < y < 5$ in sections of 0.5[mm].

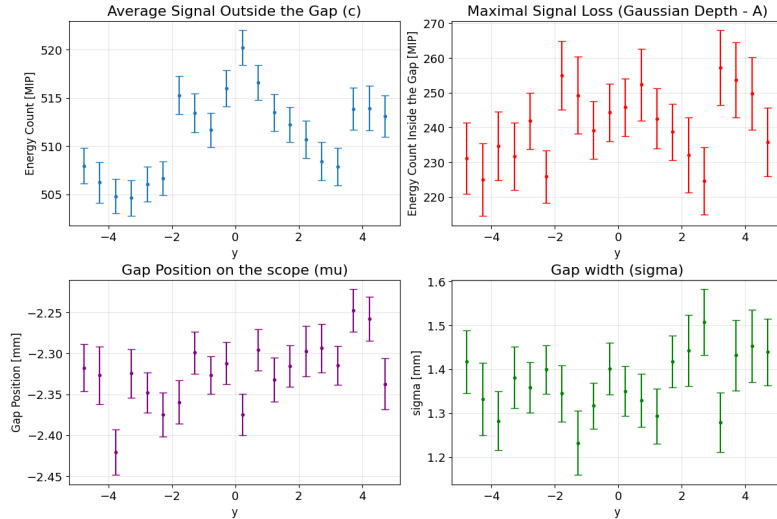


Figure 14: Run 1088: fitting parameters of the gap for varying range of y. $-5 < y < 5$, with segments of 0.5[mm]. Every value of the horizontal axis(y) is the average energies of events in this y range.

We see relatively small fluctuations in the parameters value for different y values. this means that we could take the average on events from all y to get the most accurate result.

5 Appendix: Gap Profile of Various runs

We will calculate the gap characteristics for multiple runs with 8 layers of tungsten.

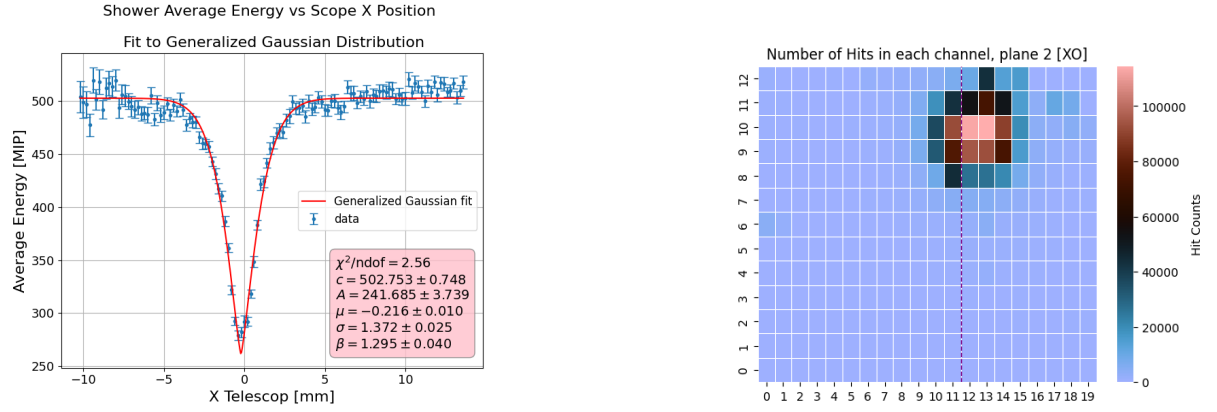


Figure 15: Run 1079

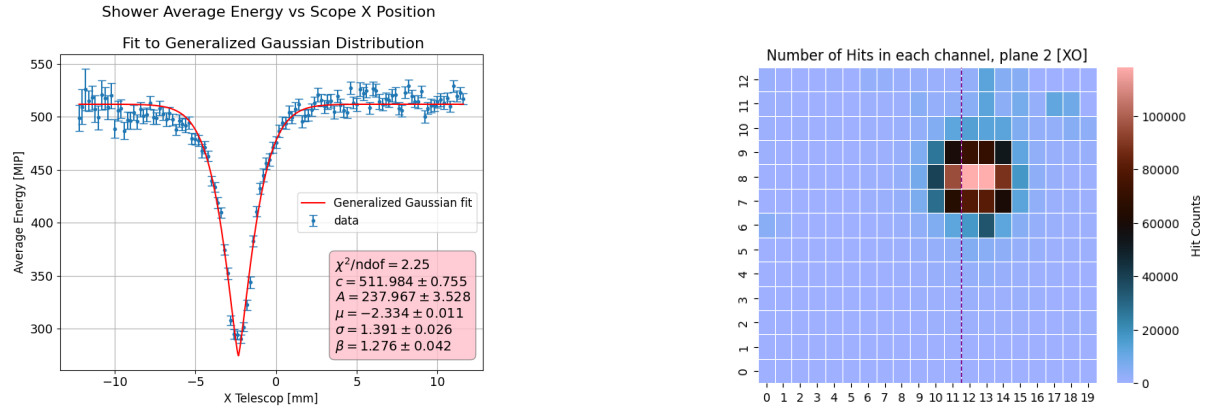


Figure 16: Run 1080

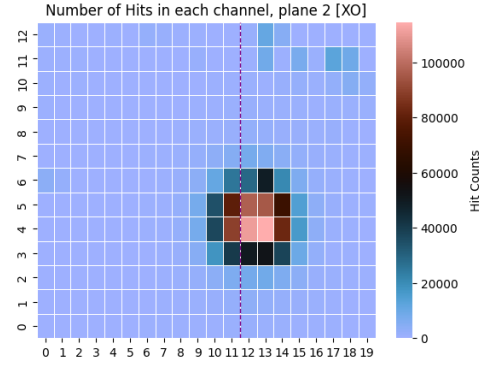
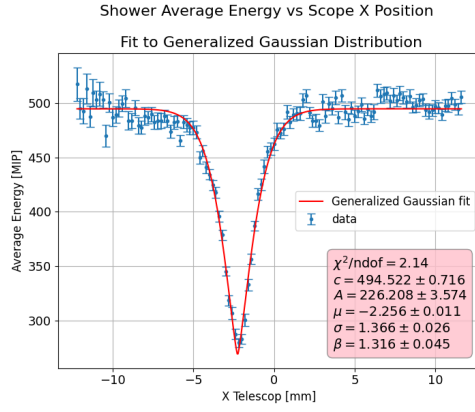


Figure 17: Run 1088

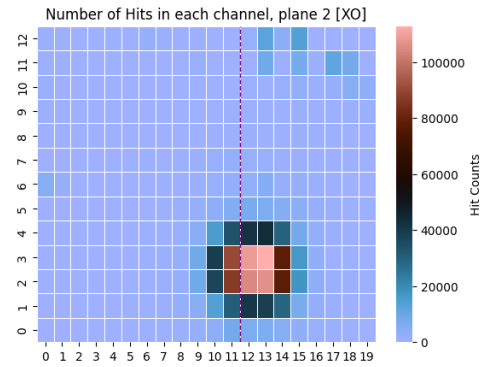
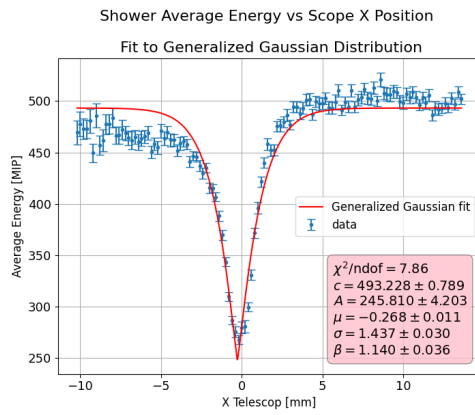


Figure 18: Run 1089

6 Notes

1. Run 1089, plane 7, ch 53, TLU 401788 we have two energy readout for the same event. amp = 232 , 162.

References

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- [3] D Strom, R Frey, M Breidenbach, D Freytag, N Graf, G Haller, O Milgrome, and V Radeka. Design and development of a dense, fine grained silicon tungsten calorimeter with integrated electronics. In *Calorimetry In Particle Physics*, pages 10–18. World Scientific, 2005.